Lecture 16 (Initialization)

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In this lecture, we will talk about initialization of the simplex method.

Initialization

Suppose we have an LP we wish to solve with the simplex method. The simplex method always assumes that you start with a BFS before applying it. If we have an LP of the form:

$$\min \ \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} \leq \vec{b}$$

$$\vec{x} \geq 0$$

we can introduce slack variables in vector $\vec{s}$ to get the standard form of the LP:

$$\min \ \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} + \vec{s} = \vec{b}$$

$$\vec{x}, \vec{s} \geq 0$$

We then can take $\vec{x} = 0, \vec{s} = \vec{b}$ in order to get a BFS to start the simplex method with. For more general LPs, it isn't so simple. If we have constraints of the form, $A\vec{x} = \vec{b}$, we cannot quickly get an initial BFS. As such, we need bigger tools. There are two common approaches to solving this issue:

1. Two Phase Simplex Method
2. Big M Method

Two Phase Simplex Method

For the **Two Phase Method**, we create a new LP to solve with the simplex method. This LP is called the phase one LP. Phase two is using the solution from the phase one LP as the initial BFS to the original LP. Suppose our original LP is in the form:

$$\min \ \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

Then the phase one LP is

$$\min \sum_{i=1}^{m} y_i$$

s.t.

$$A\vec{x} + \vec{y} = \vec{b}$$

$$\vec{x} \geq 0, \vec{y} \geq 0$$

We initialize the problem with $\vec{y} = \vec{b}$ and $\vec{x} = 0$ then apply simplex method. If the resulting solution has a positive optimal value, then the original LP is infeasible (since any positive y_i in the optimal solution means that the objective cannot be further minimized by introducing some $\vec{x}_j$ into the basis instead of leaving y_i in the basis). If instead, we find an optimal value of 0 to the phase one LP, then we take the corresponding $\vec{x}$ in the optimal solution to be the initial BFS to the original LP. Then, we can apply the simplex method once more to solve the original LP.

Notes

1) If you have constraints of the form $A_1\vec{x} \leq \vec{b}_1$ or $A_2\vec{x} \geq \vec{b}_1$, it is assumed that we can also add/subtract slack variables to get equality constraints before initialization. This will be done by first converting the LP to standard form before applying phase one.

2) You may have noticed it is possible for some entries of $\vec{b}$ to be negative. This is okay since we can multiply an entire constraint by -1 to ensure each entry of $\vec{b}$ is non-negative before setting $\vec{y} = \vec{b}$. For convenience, we assume $\vec{b} \geq 0$, but in practice you may need one preprocessing step to ensure this.

3) It is possible to have $y_i = 0$ in the basis when going through the phase one simplex method. If we have $m \geq n$ constraints (i.e. $\text{Rank}\,(A) = m$), we simply select some $x_i = 0$ to enter the basis in place of y_i when initializing. If $\text{Rank}\,(A) < n$, we can't have a BFS to the original LP, so the point is moot.

An Example

I will give an example of the setup below. Consider the following LP:

$$\max\ 5x_1 - x_2$$

s.t.

$$x_1 + x_2 \leq 5$$

$$2x_1 + 3x_2 \geq -6$$

$$x_2 \geq 2$$

$$x_1 \in \mathbb{R}, x_2 \geq 0$$

First, we will convert this into standard form. We convert the objective function to a minimize function:

$$\min\ 5x_1 - x_2$$

s.t.

$$x_1 + x_2 \leq 5$$

$$2x_1 + 3x_2 \geq -6$$

$$x_2 \geq 2$$

$$x_1 \in \mathbb{R}, x_2 \geq 0$$

We convert into the difference of two non-negative variables $x_1 - x_3 \to x_1$:

$$\min \ 5x_1 - 5x_3 - x_2$$

s.t.

$$x_1 - x_3 + x_2 \leq 5$$

$$2x_1 - 2x_3 + 3x_2 \geq -6$$

$$x_2 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

We add/subtract the appropriate slack variables, x_4, x_5, x_6:

$$\min \ 5x_1 - 5x_3 - x_2$$

s.t.

$$x_1 - x_3 + x_2 + x_4 = 5$$

$$2x_1 - 2x_3 + 3x_2 - x_5 = -6$$

$$x_2 - x_6 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Finally, we ensure $\vec{b} \geq 0$ by multiplying each constraint by -1 as needed:

$$\min \ 5x_1 - 5x_3 - x_2$$

s.t.

$$x_1 - x_3 + x_2 + x_4 = 5$$

$$-2x_1 + 2x_3 - 3x_2 + x_5 = 6$$

$$x_2 - x_6 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

So, our Phase One LP is

$$\min \ y_1 + y_2 + y_3$$

s.t.

$$x_1 - x_3 + x_2 + x_4 + y_1 = 5$$

$$-2x_1 + 2x_3 - 3x_2 + x_5 + y_2 = 6$$

$$x_2 - x_6 + y_3 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6, y_1, y_2, y_3 \geq 0$$

and we initialize phase one with $y_1 = 5$, $y_2 = 6$, $y_3 = 2$.

Big M Method

Instead of solving two LPs, the **Big M Method** allows us to solve a single LP and gives us a means to initialize. We are going to treat $M \gg 0$ as a sufficiently large number such that

$$a < \epsilon M$$

for all $a \in \mathbb{R}, \epsilon > 0$. In practice, we just choose M to be large enough to comfortably bound all of the numbers we care about. (If you want to make M rigorous, you can study ordinals, surreal, or hyperreal numbers. There are multiple ways to do so, but the theory is rather involved). For arithmetic purposes, we treat M like we would treat x in a polynomial. Some example operations are shown below:

$$3M - 2 + 7M = 10M - 2$$

$$(2 + M)(1 - M) = 2 - 2M + M - M^2 = 2 - M - M^2$$

In fact, M allows for a well defined total order: We note that

$$-(10)^{-10} M^2 \ll -90M < -90M + 1 < -M + (10)^{10} < -4 < 0 < \left((10)^{10}\right)^{10} < 0.01M < M^2$$

The largest power of M always dominates. If the coefficient of this largest power is the same (when comparing), you look at the next smallest coefficient. In fact, this ordering on M is very similar to Lexicographic Ordering! It turns out we can use M to get our initialization. Suppose we have an LP in standard form (with the same assumptions as given to the two phase method):

$$\min \ \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

Then we can get a new LP in the form:

$$\min \ \vec{c} \cdot \vec{x} + M \sum_{i=1}^{m} y_i$$

s.t.

$$A\vec{x} + \vec{y} = \vec{b}$$

$$\vec{x}, \vec{y} \geq 0$$

As $My_i \gg \vec{c} \cdot \vec{x}$ for all $\vec{x} \geq 0$, then no y_i will be in the optimal solution to the LP, if the original LP has a solution. As with two phase simplex method, we will initialize with $\vec{x} = 0$, $\vec{y} = \vec{b}$. If the optimal solution has a non-zero y_i in the basis, then the LP is infeasible. Otherwise, the solution to the Big M LP is the optimal solution to the original LP.

Notes

1) In practice, picking the appropriate M is important. If too small of an M is chosen, then sometimes the resulting solution will yield incorrect results. If M is too large, you may get rounding errors in practice. If you implement M exactly as a total order (which is what we assume when doing theory), the simplex method will run slower, but the errors above won't be an issue. The selection of $<$ should depend on the problem when implementing in practice.

An Example

Consider the same example as we saw for the Two Phase Method:

$$\max\ 5x_1 - x_2$$

s.t.

$$x_1 + x_2 \leq 5$$

$$2x_1 + 3x_2 \geq -6$$

$$x_2 \geq 2$$

$$x_1 \in \mathbb{R}, x_2 \geq 0$$

Like the Two Phase Method, we convert this into the appropriate standard form first:

$$\min\ 5x_1 - 5x_3 - x_2$$

s.t.

$$x_1 - x_3 + x_2 + x_4 = 5$$

$$-2x_1 + 2x_3 - 3x_2 + x_5 = 6$$

$$x_2 - x_6 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Now, we get our Big M formulation:

$$\min\ 5x_1 - 5x_3 - x_2 + M\left(y_1 + y_2 + y_3\right)$$

s.t.

$$x_1 - x_3 + x_2 + x_4 + y_1 = 5$$

$$-2x_1 + 2x_3 - 3x_2 + x_5 + y_2 = 6$$

$$x_2 - x_6 + y_3 = 2$$

$$x_1, x_2, x_3, x_4, x_5, x_6, y_1, y_2, y_3 \geq 0$$

And we initialize with $y_1 = 5$, $y_2 = 6$, $y_3 = 2$.